

Attention to syntax and semantics

Meaning matters

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Abstract

We introduce a new attention mechanism for both syntax and semantics. An implicit deep learning model is used to jointly learn Q-K-V embeddings and contextual token representations designed to simultaneously capture syntactic co-occurrence and semantic alignment.

Jointly learning LLM semantics and syntax

Chunk a corpus into data sets of size N
(typically large to capture language meaning).

For a chunk with input tokens $\{t_1, \dots, t_N\} \subset \mathbb{R}^r$
and corresponding outputs $\{y_1, \dots, y_N\} \subset \mathbb{R}^q$
use mini-batch SGD to learn matrices

K, Q, V, Q' and fixed points $s_i \in \mathbb{R}^n$, $i \in [N]$

so that:

$$s_i = \varphi \left(\underset{n \times n}{K} s_i + \underset{n \times r}{Q} t_i \right)$$

$$y_i = \underset{q \times n}{V} s_i + \underset{q \times r}{Q'} t_i$$

We now specify an implicit deep learning model [1]
to jointly learn language semantics and syntax.

Let input matrix $U = [t_1, \dots, t_N] \in \mathbb{R}^{r \times N}$

and output matrix $Y = [\varepsilon, \dots, \varepsilon] \in \mathbb{R}^{q \times N}$

where $\varepsilon \in \mathbb{R}^q$ is a hyperparameter

and we choose $q = n$, for reasons to be revealed later.

(On a few sample batches of N one can apply

· Nesterov Rademacher descent [2] to find an ε

whose implicit model has reduced average Renyi α -entropy.)

For implicit model:

$$X = \varphi(KX + QU) \in \mathbb{R}^{n \times N}$$

$$\hat{Y} = VX + Q'U$$

where $K = \mathcal{J} - \text{diag}(|\mathcal{J}| \mathbb{I}_n) + \gamma \mathbb{I}_n$, with

hyperparameter $0 < \gamma < 1$, ensures

K is well-posed for φ .

Our training problem to jointly learn language semantics and syntax is:

$$\min_{\theta, K, Q, V, Q', X} \left\{ \mathcal{L}(Y, VX + Q'U) + \lambda_1 \mathcal{P}_1(Q', K) + \lambda_2 \mathcal{P}_2(\theta, Q', K, V) \right\}$$

Penalty $\mathcal{P}_1$ attends to semantics and

penalty $\mathcal{P}_2$ attends to syntax and

$\mathcal{L}$ is a loss function.

For mini-batch SGD, let batch size $1 < B < N$.

We now go on to define penalties $\mathcal{P}_1$ and $\mathcal{P}_2$.

Let $q_i = Qt_i$, $k_i = Ks_i$

$q'_i = Q't_i$, $v_i = Vs_i$

For $i \in [B]$, define probabilities

$$\alpha_{ij} = \underset{j \in [B]}{\text{softmax score}}(q'_i, k_j)$$

where score is a similarity function e.g. a kernel

$$(\dim q'_i = q = n = \dim k_j)$$

$$\text{Define penalty } \mathcal{P}_i(Q', K) = \frac{1}{B} \sum_{i=1}^B H_\alpha^i$$

where $H_\alpha^i \triangleq H_\alpha(\{\alpha_{ij}\}_{j=1}^B)$, the Renyi α -entropy
of distribution $\{\alpha_{ij}\}_{j=1}^B$.

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Let contextual representation of token t_i be:

$$h_i = \sum_{j=1}^B \alpha_{ij} v_j \in \mathbb{R}^n$$

For L layers, let parameters

$$\Theta = (W_L, \dots, W_1), \quad W_\ell \in \mathbb{R}^{n \times n}$$

Let $h_i^0 = h_i \in \mathbb{R}^n$

Define update $h_i^L = \text{ReLU}(W_L h_i^{L-1})$, $L \in [L]$

For $i \in [B]$, define probabilities

$$\beta_{ij} = \underset{j \in [B]}{\text{softmax}} \text{score}(h_i^L, k_j)$$

and let $j_i^\beta = \arg \max_{j \in [B]} \beta_{ij}$.

Define output token $o_{i+1} = t_{j_i^\beta} \in \mathbb{R}^r$

corresponding to input token t_i .

Define penalty $\mathcal{P}_2(\theta, Q', K, V) = \text{Perplexity}(o_2, \dots, o_{B+1})$
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After mini-batch SGD minimization on a chunk,
 we will have jointly learned embeddings K, Q, V, Q'
 and fixed points X and FeedForward network parameters θ ,
 as well as for each token t in the chunk an output
 token o .

Summary

We have specified an implicit deep learning model to jointly learn language semantics and syntax, in the sense that syntactic co-occurrence is quantified by perplexity and semantic alignment by Rényi α -entropy. On each corpus chunk we use mini-batch SGD to jointly learn Q-K-V embeddings and feedforward network parameters. Given a token from a prompt, we look up the chunk it resides in and retrieve its corresponding pre-computed output token. An attention mechanism for both syntax and semantics.

□

References

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